Federated Learning for IoT Intrusion Detection

A Controlled Evaluation of Centralized Training, FedAvg, and FedProx under IID and Non-IID Data

---

Research-grade experimental report

Based on six executed notebooks supplied for the project

Dataset:	CICIoT2023 (Kaggle mirror used in execution)
Cleaned samples:	45,018,243
Training / test:	36,014,594 / 9,003,649
Input / output:	39 features / 34 classes
Model:	MLP 39–128–64–34 (15,586 trainable parameters)
Compute:	NVIDIA Tesla T4, CUDA, mixed precision where available

October 2026

Abstract

This report analyzes a controlled set of federated intrusion-detection experiments performed on the full cleaned CICIoT2023 tabular dataset. The experiments compare a centralized multilayer perceptron baseline with FedAvg under an IID client partition, FedAvg under severe Dirichlet label skew, and FedProx under the exact same non-IID partition. The study controls the dataset, train/test split, model architecture, optimizer, batch size, learning rate, random seed, training exposure, and global test set. This makes the observed differences primarily attributable to federation, client heterogeneity, and the FedProx local objective.

The central result is that federation itself caused only a modest reduction under IID partitioning, whereas non-IID label skew caused a substantially larger loss in class-balanced performance. Centralized Macro F1 was 56.70%, FedAvg-IID reached 52.90%, FedAvg-Non-IID fell to 28.80%, and FedProx-Non-IID with $\mu = 0.01$ reached 23.36%. Under the tested three-round, one-local-epoch budget, FedProx did not improve the non-IID result. The report therefore treats the FedProx outcome as a configuration-specific negative result rather than a general statement about the algorithm. Detailed per-class metrics, convergence histories, partition statistics, runtime measurements, and confusion matrices are included.

Contents

1 Study Scope and Research Questions **1**

2 Background **1**

2.1 Federated learning and FedAvg 1

2.2 FedProx 1

2.3 Dataset context 2

3 Data Preparation and Partitioning **2**

3.1 Loaded data and cleaning 2

3.2 Severe class imbalance 2

3.3 IID client partition 3

3.4 Non-IID client partition 3

4 Experimental Methodology **5**

4.1 Controlled design 5

4.2 Model 6

4.3 Training-budget matching 6

4.4 Evaluation metrics 6

5 Results **7**

5.1 Headline comparison 7

5.2 Isolating the three experimental effects 8

5.3 Convergence behavior 9

5.4 Client-level dispersion under non-IID data 9

5.5 Class-level behavior 10

5.6 FedProx class-level effect . . . 12
5.7 Runtime . . . 13

6 Discussion **13**
6.1 Federation is not the main source of loss in this experiment . . . 13
6.2 Non-IID label skew is the dominant challenge . . . 14
6.3 Why the tested FedProx configuration may not help . . . 14
6.4 Class imbalance is visible across every training regime . . . 14

7 Threats to Validity and Limitations **14**

8 Reproducibility and Experimental Artifacts **15**

9 Recommended Next Experiments **16**

10 Conclusion **16**

A Full Class Distribution **17**

B Federated Round Histories **17**
B.1 FedAvg-IID . . . 18
B.2 FedAvg-Non-IID . . . 18
B.3 FedProx-Non-IID . . . 18

C Full Per-Class Precision, Recall, and F1 **18**
C.1 Centralized . . . 18
C.2 FedAvg-IID . . . 19
C.3 FedAvg-Non-IID . . . 20
C.4 FedProx-Non-IID . . . 21

D Normalized Confusion Matrices **22**

List of Figures

1 Full class distribution after cleaning. A logarithmic horizontal axis is used because class counts span more than three orders of magnitude. . . . 3
2 Client dataset sizes. IID partitioning produces nearly equal clients; the Dirichlet partition creates strong quantity skew in addition to label skew. . . . 4
3 Within-client class proportions for the Dirichlet non-IID partition (direct notebook output). Concentrated bands indicate that different clients are dominated by different attack labels. . . . 4
4 Controlled experimental design reconstructed from the supplied notebooks. . . . 5

5 Headline test metrics. The gap between weighted and macro metrics becomes much larger under non-IID training, indicating that performance is concentrated in a subset of classes. 7
6 Measured effect sizes from the controlled comparisons. 8
7 Global test performance after each federated round. 9
8 Weighted local training loss. The FedProx objective includes the proximal penalty; its cross-entropy component is plotted separately for comparison. 9
9 Per-class F1 scores for all four experiments. 10
10 Number of classes with no successful F1 performance on the shared test set. . . . 11
11 Per-class F1 change when FedProx replaces FedAvg on the exact same non-IID partition. 12
12 Recorded training times. FedProx local training is slower because each batch additionally evaluates the proximal penalty over model parameters. 13
13 Centralized baseline normalized confusion matrix (direct notebook output). . . . 24
14 FedAvg-IID normalized confusion matrix (direct notebook output). 26
15 FedAvg-Non-IID normalized confusion matrix (direct notebook output). 28
16 FedProx-Non-IID normalized confusion matrix (direct notebook output). 30

List of Tables

1 Non-IID client heterogeneity summary. 5
2 Common configuration and intentional differences. 6
3 Final performance on the shared global test set. Percentage metrics are shown in percent. 7
4 Changes attributable to each controlled comparison. All entries are percentage-point changes; MCC is multiplied by 100 for the same scale. Negative values denote lower performance in the second condition. 8
5 Training time reported by the notebooks. These values describe the recorded Tesla T4 runs; they do not include all data loading and preprocessing time. 13
6 Class counts in the cleaned 45,018,243-row dataset. 17
7 Per-class metrics: Centralized. 18
8 Per-class metrics: FedAvg-IID. 19
9 Per-class metrics: FedAvg-Non-IID. 20
10 Per-class metrics: FedProx-Non-IID. 21

1 Study Scope and Research Questions

The project asks a practical research question: how does a compact intrusion-detection model behave when the same large IoT traffic dataset is moved from centralized training to federated learning, and how strongly does client heterogeneity alter that behavior?

The supplied notebooks implement four directly comparable experiments:

1. **Centralized baseline**: all training samples are optimized in one process for three epochs.
2. **FedAvg-IID**: the same training split is divided across 10 nearly identical IID clients; all clients participate in three federated rounds with one local epoch per round.
3. **FedAvg-Non-IID**: the same training split is repartitioned with Dirichlet label skew ($\alpha = 0.1$), keeping the remaining training settings unchanged.
4. **FedProx-Non-IID**: the exact same non-IID client partition is reused, with the local objective changed to FedProx and $\mu = 0.01$.

The analysis is organized around three experimental effects:

- **Federation effect**: FedAvg-IID minus Centralized.
- **Heterogeneity effect**: FedAvg-Non-IID minus FedAvg-IID.
- **FedProx effect**: FedProx-Non-IID minus FedAvg-Non-IID.

All reported numerical results are taken from the executed notebook outputs. No accuracy, F1 score, or runtime value is re-estimated from synthetic data.

2 Background

2.1 Federated learning and FedAvg

Federated learning keeps local training data at participating clients and aggregates model updates into a shared global model. The FedAvg algorithm introduced iterative model averaging as a practical federated optimization method [2]. If client k returns parameters w_k^{t+1} after local training on n_k examples, the server update used in these notebooks is sample-weighted averaging:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{\sum_j n_j} w_k^{t+1}.$$

In the experiments, all 10 clients participate in every round. Therefore the comparison does not include partial participation, client dropout, stragglers, or communication failure.

2.2 FedProx

FedProx modifies local client optimization to discourage large deviations from the current global model [3]. In the supplied implementation, each client minimizes:

$$\mathcal{L}_k^{\text{prox}}(w) = \mathcal{L}_k^{\text{CE}}(w) + \frac{\mu}{2} \left\| w - w^t \right\|_2^2,$$

with $\mu = 0.01$. The aggregation step remains sample-weighted averaging. Thus the algorithmic difference from FedAvg is isolated to local training.

2.3 Dataset context

CICIoT2023 was introduced as a large-scale IoT intrusion-detection dataset generated from a topology of 105 devices, with 33 attacks grouped into seven attack categories [1]. The experiments here use a fine-grained 34-class target space: 33 attack labels plus BENIGN. The notebooks report 39 input features.

3 Data Preparation and Partitioning

3.1 Loaded data and cleaning

The preparation notebook loads 63 CSV shards from the Kaggle mirror `mashrurfaiyaz28/ciot2023-2`. It reads 45,019,243 raw rows and retains 45,018,243 after replacing infinite values with missing values and dropping rows containing missing values. Exactly 1,000 rows are removed.

A deterministic stratified 80/20 split with seed 42 produces:

- 36,014,594 training samples,
- 9,003,649 test samples,
- 39 numeric features,
- 34 classes.

StandardScaler statistics are fit on the *entire training split* and then reused in all experiments. This is an appropriate control for algorithm comparison, but it also means the simulation assumes globally computed preprocessing statistics before federation. That design choice is revisited as a privacy/realisim limitation in Section 7.

3.2 Severe class imbalance

The cleaned dataset is highly imbalanced. The largest class, DDOS-ICMP-FLOOD, contains 6,893,196 samples, while UPLOADING-ATTACK contains only 1,196. The largest-to-smallest count ratio is approximately 5,763.5:1. The five most frequent labels together account for 53.73% of all cleaned samples, whereas the ten smallest labels account for only 0.197%.

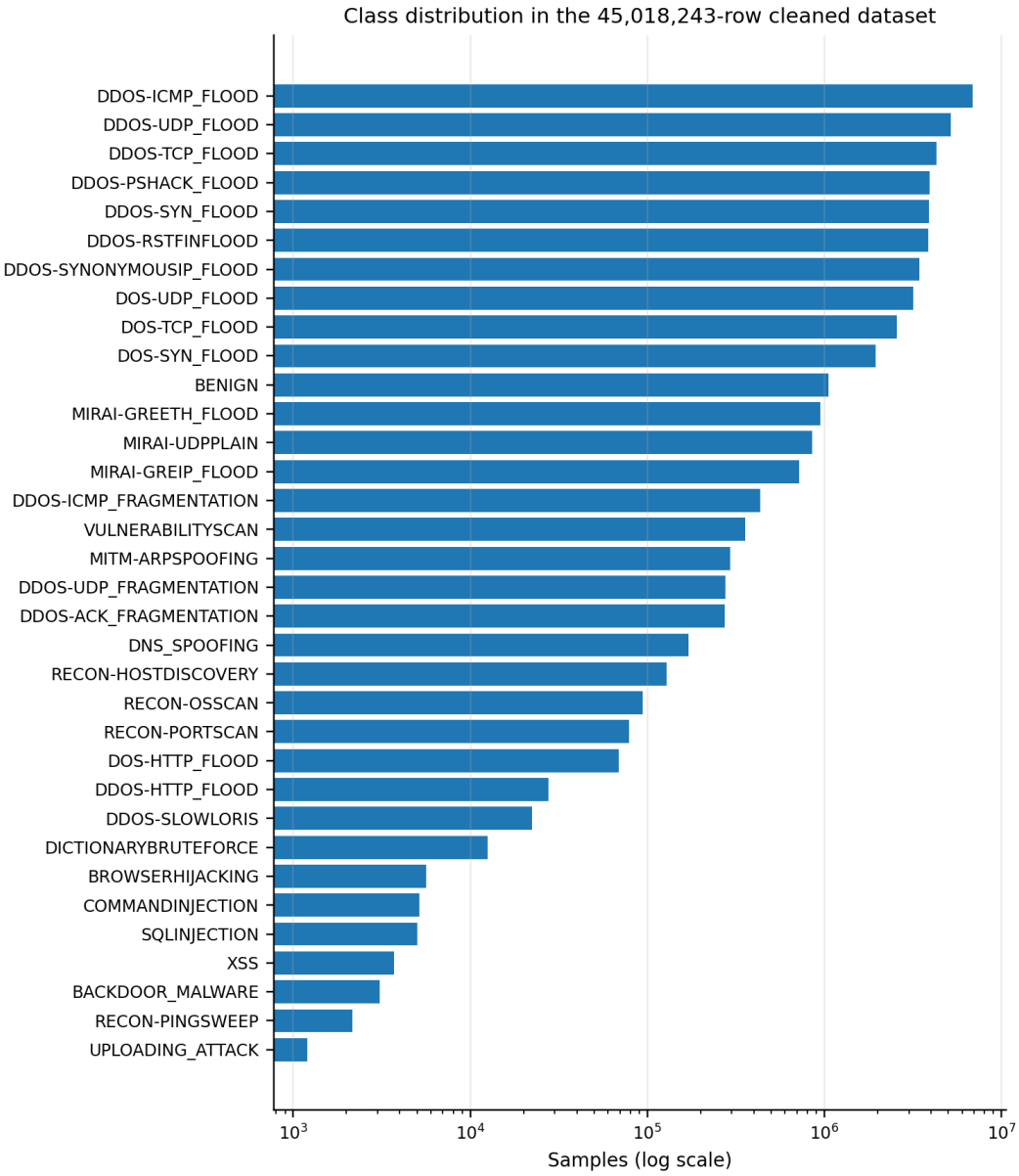


Figure 1: Full class distribution after cleaning. A logarithmic horizontal axis is used because class counts span more than three orders of magnitude.

This imbalance makes raw accuracy insufficient as the only outcome measure. Macro F1, balanced accuracy, MCC, and class-level recall are essential for seeing failures on minority classes.

3.3 IID client partition

For the IID experiment, each class is independently shuffled and split across 10 clients. This produces almost perfectly matched client distributions. Client sizes range only from 3,601,446 to 3,601,477 samples, a maximum difference of 31 rows. The largest same-class percentage spread across clients is 0.000104 percentage points.

3.4 Non-IID client partition

For the heterogeneous experiments, the same 36,014,594 training rows are repartitioned with a deterministic Dirichlet label-skew procedure using $\alpha = 0.1$. Every training sample is assigned

exactly once; no row is dropped or duplicated. The resulting client sizes range from 902,858 to 9,168,171 samples, a 10.15× size ratio and a size coefficient of variation of 0.6814.

Across the 10 clients, the mean number of active classes is 23.6 out of 34, the mean dominant-class share is 59.28%, and the mean Jensen-Shannon divergence from the global label distribution is 0.3767. These statistics confirm that the non-IID setup introduces substantial label and quantity heterogeneity.

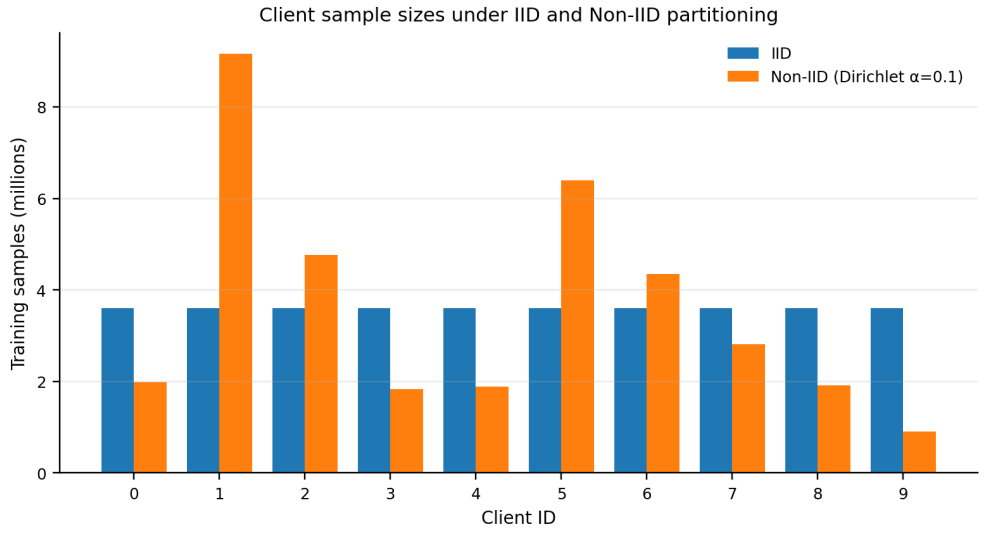


Figure 2: Client dataset sizes. IID partitioning produces nearly equal clients; the Dirichlet partition creates strong quantity skew in addition to label skew.

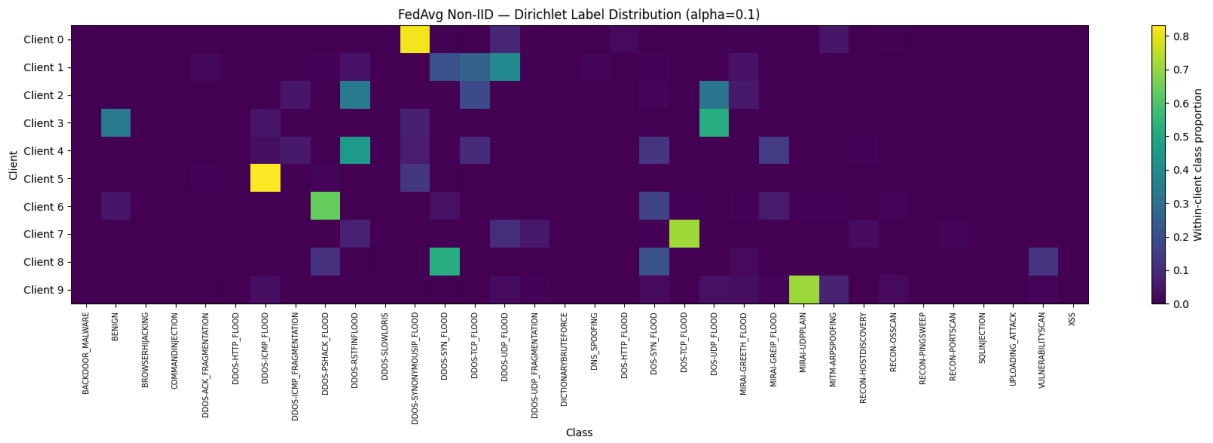


Figure 3: Within-client class proportions for the Dirichlet non-IID partition (direct notebook output). Concentrated bands indicate that different clients are dominated by different attack labels.

Table 1: Non-IID client heterogeneity summary.

Client	Samples	Share (%)	Active	Dominant class	Dominant (%)	Norm. entropy	JS div.
0	1,980,544	5.50	22	DDOS-SYNONYMOUSIP_FLOOD	81.49	0.2095	0.4534
1	9,168,171	25.46	26	DDOS-UDP_FLOOD	39.39	0.4481	0.2579
2	4,766,651	13.24	24	DDOS-RSTFINFLOOD	34.10	0.4433	0.3159
3	1,829,373	5.08	24	DOS-UDP_FLOOD	52.07	0.3239	0.4072
4	1,884,406	5.23	22	DDOS-RSTFINFLOOD	45.35	0.4806	0.3131
5	6,400,792	17.77	24	DDOS-ICMP_FLOOD	83.17	0.1682	0.3697
6	4,354,291	12.09	27	DDOS-PSHACK_FLOOD	63.37	0.3614	0.3799
7	2,817,647	7.82	21	DOS-TCP_FLOOD	71.18	0.3002	0.4230
8	1,909,861	5.30	18	DDOS-SYN_FLOOD	51.72	0.3743	0.3986
9	902,858	2.51	28	MIRAI-UDPPLAIN	70.95	0.3577	0.4484

4 Experimental Methodology

4.1 Controlled design

Figure 4 summarizes the experiment structure. The comparison notebook explicitly verifies that dataset fraction, train/test sizes, feature count, class count, architecture, random seed, batch size, learning rate, and effective data passes are identical across all four experiments. It also verifies that FedAvg-Non-IID and FedProx-Non-IID use the exact same client-position arrays.

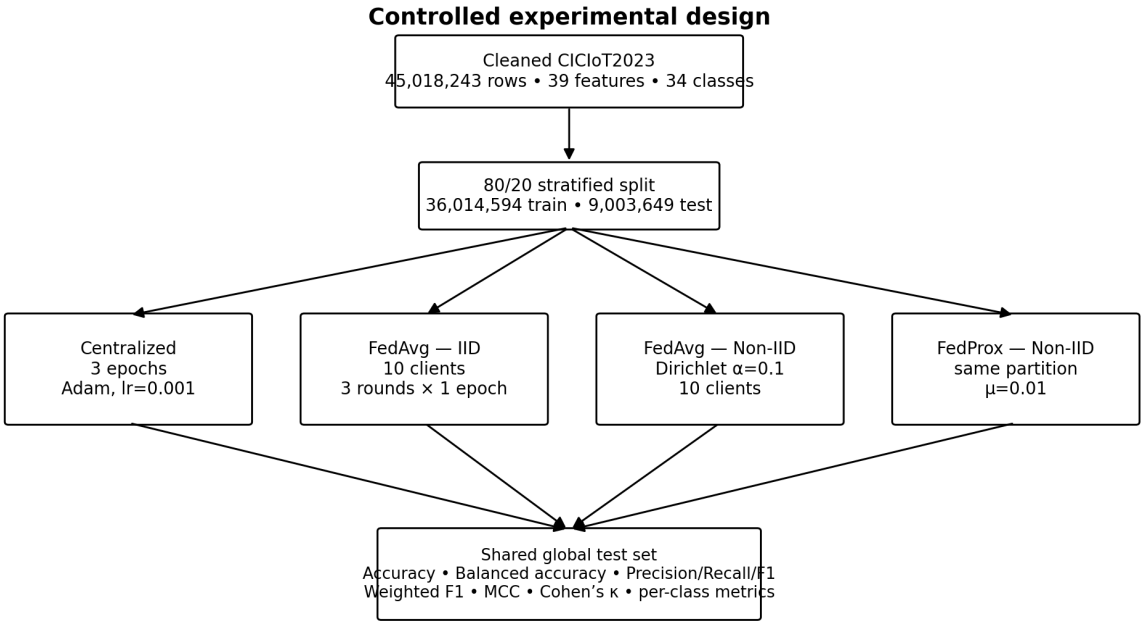


Figure 4: Controlled experimental design reconstructed from the supplied notebooks.

Table 2: Common configuration and intentional differences.

Setting	Centralized	FedAvg-IID	FedAvg-Non-IID	FedProx-Non-IID
Dataset fraction	1.00	1.00	1.00	1.00
Train / test	36,014,594 / 9,003,649	same	same	same
Features / classes	39 / 34	39 / 34	39 / 34	39 / 34
Model	39–128–64–34	same	same	same
Trainable params	15,586	15,586	15,586	15,586
Optimizer	Adam	Adam	Adam	Adam
Learning rate	0.001	0.001	0.001	0.001
Batch size	8192	8192	8192	8192
Class weights	off	off	off	off
Training exposure	3 epochs	3 rounds × 1 epoch	3 rounds × 1 epoch	3 rounds × 1 epoch
Clients	–	10	10	10
Participation	–	all clients/round	all clients/round	all clients/round
Partition	centralized	IID stratified	Dirichlet $\alpha = 0.1$	same exact non-IID split
FedProx μ	–	–	–	0.01
Compute	Tesla T4	Tesla T4	Tesla T4	Tesla T4
AMP	on	on	on	on

4.2 Model

All experiments use the same multilayer perceptron:

$$39 \to 128 \to 64 \to 34,$$

with ReLU activations after the two hidden layers and 15,586 trainable parameters. Cross-entropy loss is used without class weights. Adam is re-created for each local client training call in the federated notebooks.

4.3 Training-budget matching

The centralized model receives three complete passes over the training split. In the FL experiments, all 10 clients participate in each of three rounds and train for one local epoch. Since the union of client datasets covers the full training split exactly once per round, this gives three full data passes overall. The match is not identical at the optimizer-state level: centralized Adam retains optimizer state across epochs, while each FL client starts a fresh Adam optimizer for its local training call. The report therefore treats "equal data exposure" rather than "identical optimization trajectory" as the controlled quantity.

4.4 Evaluation metrics

All models are evaluated on the same 9,003,649-sample global test set. The notebooks report:

- accuracy,
- balanced accuracy,
- macro precision, macro recall, and Macro F1,
- weighted precision, weighted recall, and Weighted F1,
- Matthews correlation coefficient (MCC),

- Cohen's κ,
- precision, recall, F1, and support for every class,
- normalized confusion matrices.

For this imbalanced 34-class problem, Macro F1 is treated as a primary comparison metric because it weights each class equally. Weighted F1 and accuracy remain useful for aggregate traffic-level performance, but they are strongly influenced by frequent attack types.

5 Results

5.1 Headline comparison

Table 3: Final performance on the shared global test set. Percentage metrics are shown in percent.

Experiment	Acc.	Bal. Acc.	Macro P	Macro R	Macro F1	Weighted F1	MCC	κ
Centralized	77.93	55.37	66.00	55.37	56.70	76.60	0.7598	0.7576
FedAvg-IID	77.16	52.57	58.84	52.57	52.90	74.99	0.7521	0.7488
FedAvg-Non-IID	65.74	36.24	30.34	36.24	28.80	57.91	0.6336	0.6235
FedProx-Non-IID	64.91	31.28	26.14	31.28	23.36	56.95	0.6251	0.6148

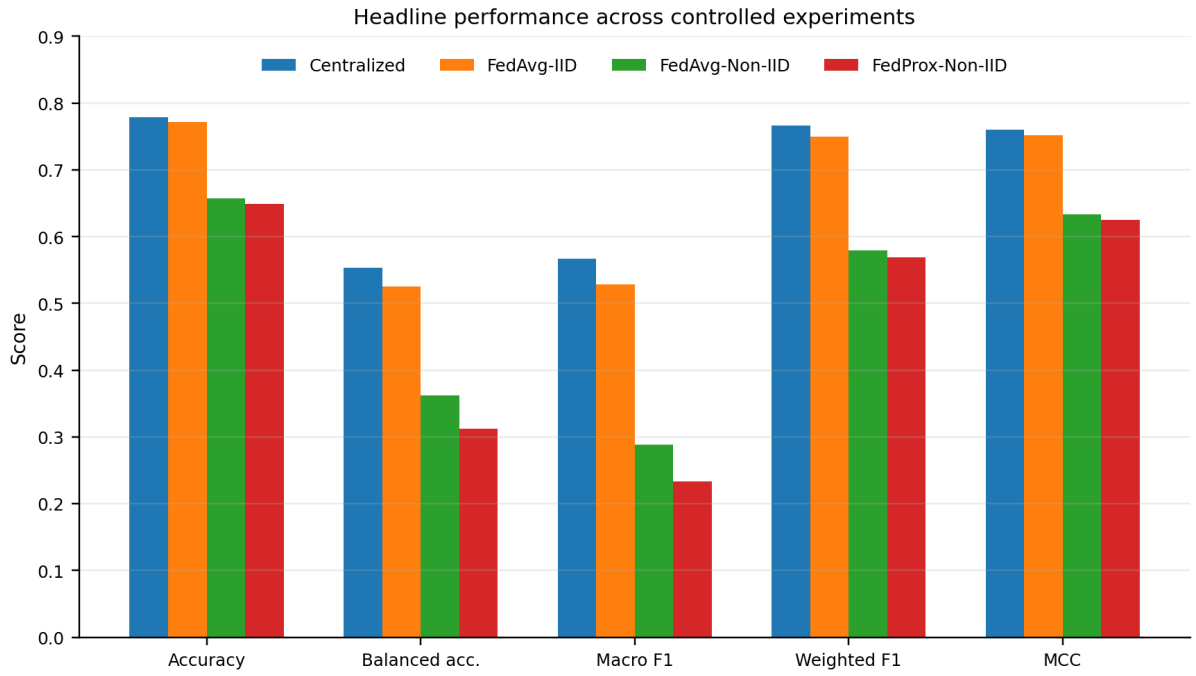


Figure 5: Headline test metrics. The gap between weighted and macro metrics becomes much larger under non-IID training, indicating that performance is concentrated in a subset of classes.

The centralized baseline achieves 77.93% accuracy and 56.70% Macro F1. FedAvg under IID partitioning reaches 77.16% accuracy and 52.90% Macro F1, close to the centralized result in aggregate accuracy but lower on class-balanced performance.

The major degradation appears when client data become non-IID. FedAvg-Non-IID falls to 65.74% accuracy and 28.80% Macro F1. FedProx-Non-IID with $\mu = 0.01$ reaches 64.91% accuracy and 23.36% Macro F1.

5.2 Isolating the three experimental effects

Table 4: Changes attributable to each controlled comparison. All entries are percentage-point changes; MCC is multiplied by 100 for the same scale. Negative values denote lower performance in the second condition.

Comparison	Accuracy	Bal. Acc.	Macro F1	Weighted F1	MCC
Federation effect	-0.774	-2.798	-3.803	-1.614	-0.769
Non-IID effect	-11.417	-16.334	-24.099	-17.073	-11.849
FedProx effect	-0.835	-4.961	-5.436	-0.966	-0.849

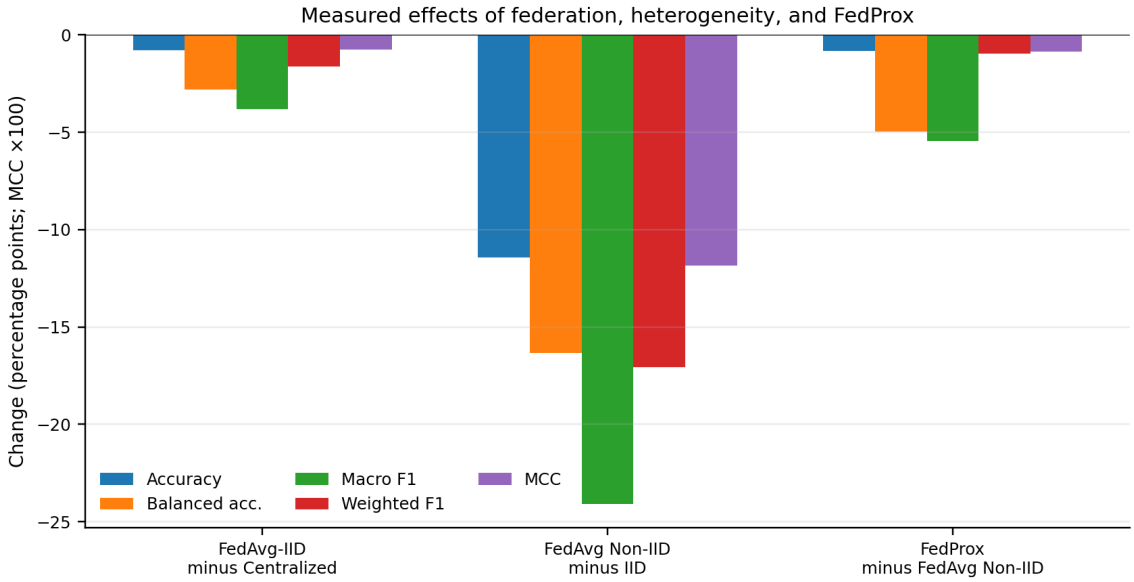


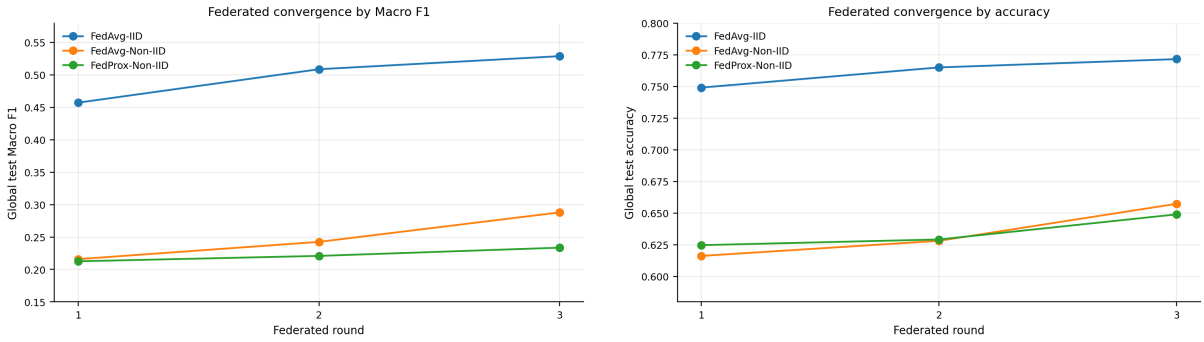
Figure 6: Measured effect sizes from the controlled comparisons.

The **federation effect under IID data** is relatively modest: accuracy decreases by 0.77 points and Macro F1 by 3.80 points compared with centralized training.

The **non-IID effect** is much larger: moving from IID to Dirichlet-skewed clients reduces accuracy by 11.42 points and Macro F1 by 24.10 points. This is the dominant measured effect in the experiment.

The **FedProx effect** is negative under the tested setting: relative to FedAvg on the exact same non-IID client split, Macro F1 decreases by 5.44 points and balanced accuracy by 4.96 points. This should be read as a result for $\mu = 0.01$, three rounds, one local epoch, Adam, and this partition. It is not evidence that FedProx is generally inferior to FedAvg.

5.3 Convergence behavior



(a) Macro F1 **(b)** Accuracy

Figure 7: Global test performance after each federated round.

FedAvg-IID improves Macro F1 from 45.73% after round 1 to 52.90% after round 3, a gain of 7.16 points. FedAvg-Non-IID also improves over the three rounds, from 21.59% to 28.80%, but remains far below the IID trajectory. FedProx improves more slowly in Macro F1, from 21.26% to 23.36%.

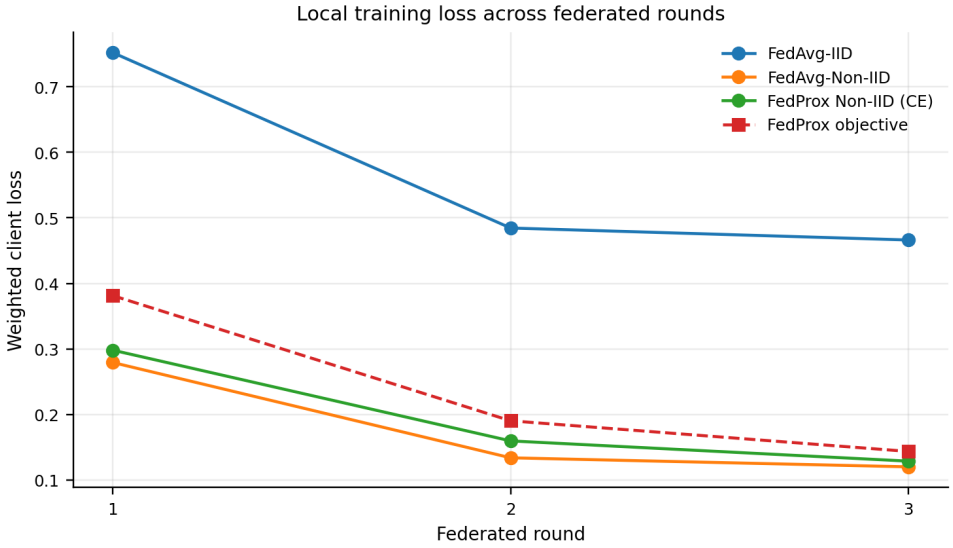


Figure 8: Weighted local training loss. The FedProx objective includes the proximal penalty; its cross-entropy component is plotted separately for comparison.

FedProx's weighted proximal penalty decreases from 0.0833 in round 1 to 0.0148 in round 3, showing that the local regularization term becomes smaller as training proceeds. However, this reduction does not translate into higher global Macro F1 within the three-round budget.

5.4 Client-level dispersion under non-IID data

Local training accuracy is high even when global class-balanced performance is poor. In FedAvg-Non-IID, weighted local training accuracy rises from 94.74% to 95.99%, while the mean client accuracy ranges around 91.90–93.62% and the client-level standard deviation remains approximately 5.8–6.7 points. FedProx shows a very similar local-accuracy dispersion. This contrast is important: highly specialized clients can fit their dominant local labels well while the

aggregated global model still performs poorly across all 34 classes.

5.5 Class-level behavior

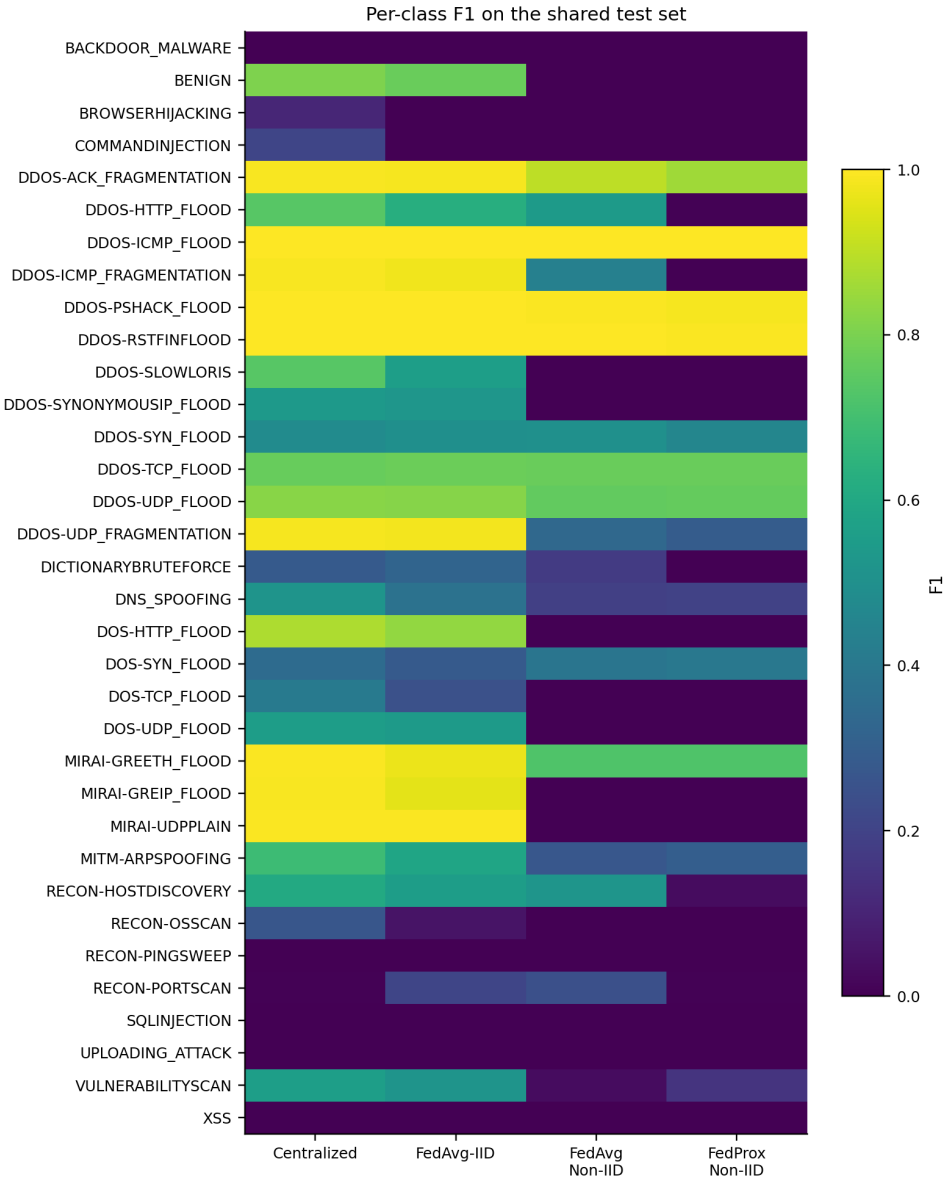


Figure 9: Per-class F1 scores for all four experiments.

The centralized and IID models retain strong F1 scores for several high-frequency flooding and Mirai classes, while rare web/recon classes remain difficult. Under non-IID training, the model becomes much less balanced across classes. The number of classes with F1 exactly zero rises from 5 in the centralized model and 7 in FedAvg-IID to 14 in both non-IID runs.

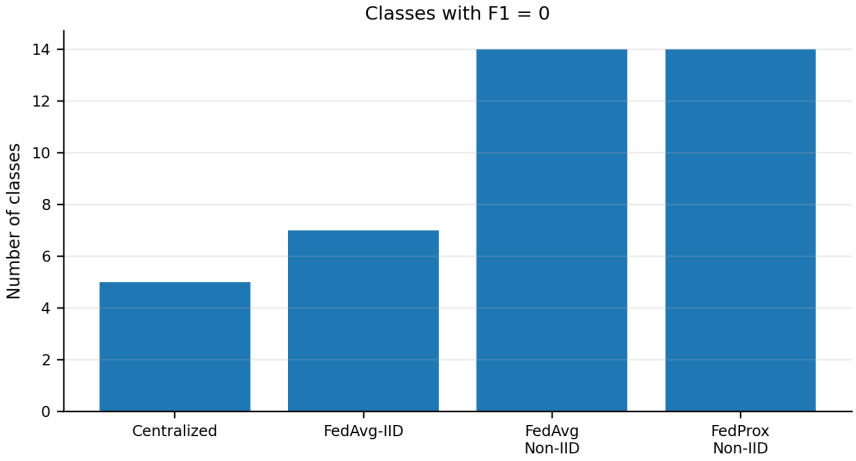


Figure 10: Number of classes with no successful F1 performance on the shared test set.

The full per-class tables in Appendix C show that the non-IID degradation is not uniform. Some dominant flooding classes remain near-perfect, while many minority or underrepresented labels collapse.

5.6 FedProx class-level effect

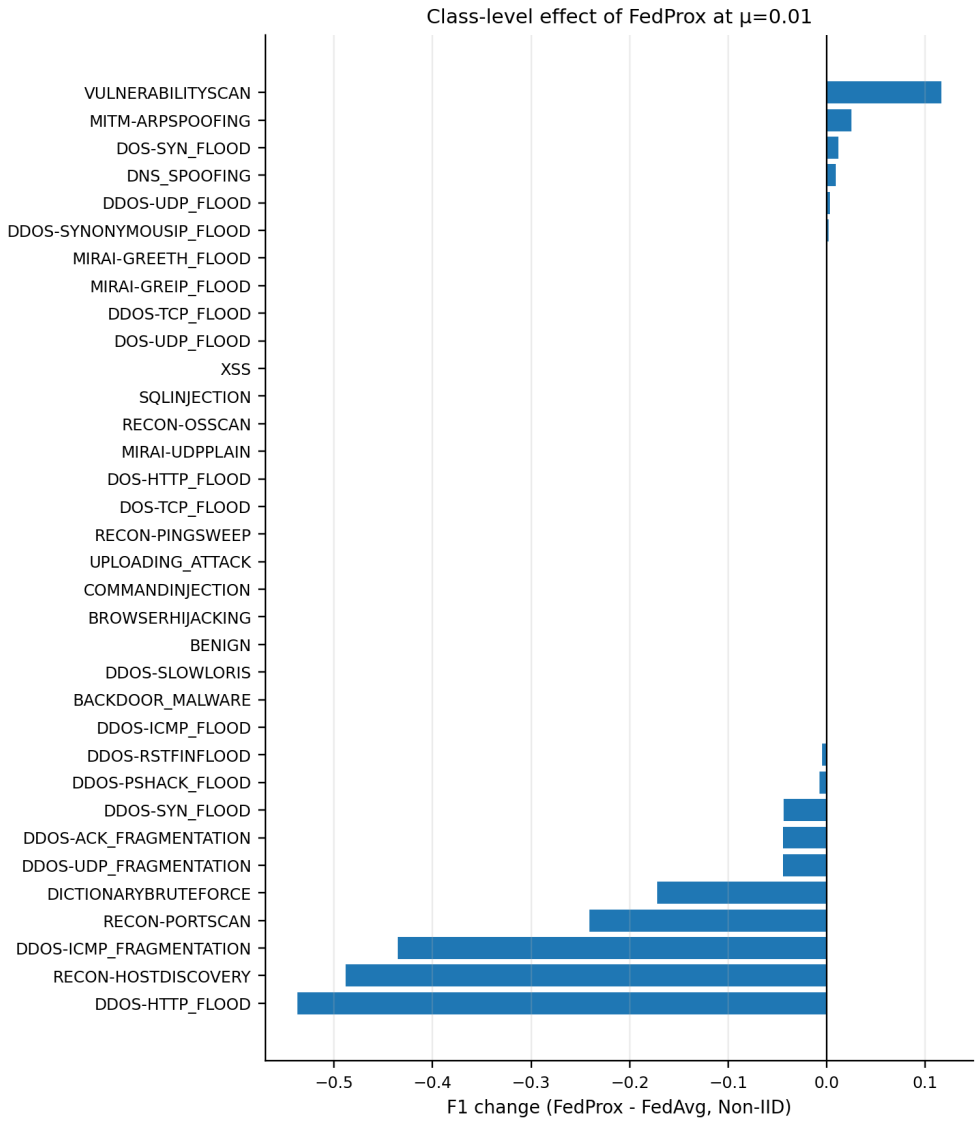


Figure 11: Per-class F1 change when FedProx replaces FedAvg on the exact same non-IID partition.

FedProx improves some classes, most notably VULNERABILITYSCAN (approximately +0.116 F1), but loses substantially on others. The largest decreases are DDOS-HTTP-FLOOD (about -0.537), RECON-HOSTDISCOVERY (-0.488), DDOS-ICMP-FRAGMENTATION (-0.435), RECON-PORTSCAN (-0.241), and DICTIONARYBRUTEFORCE (-0.172). These mixed class-level effects explain why the global Macro F1 declines even though some labels improve.

5.7 Runtime

Table 5: Training time reported by the notebooks. These values describe the recorded Tesla T4 runs; they do not include all data loading and preprocessing time.

Experiment	Training seconds	Minutes
Centralized	31.41	0.52
FedAvg-IID	37.62	0.63
FedAvg-Non-IID	36.81	0.61
FedProx-Non-IID	53.29	0.89

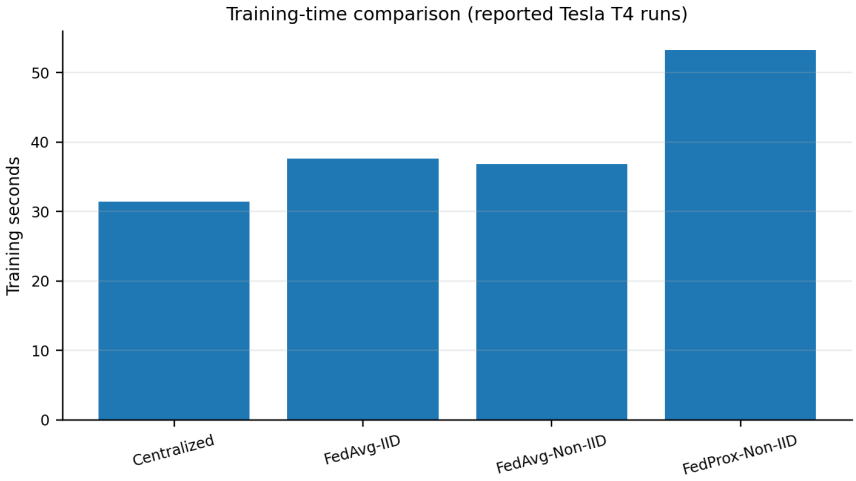


Figure 12: Recorded training times. FedProx local training is slower because each batch additionally evaluates the proximal penalty over model parameters.

FedProx's measured round-training time is approximately 44.8% higher than FedAvg under the same non-IID split. Because the notebooks were executed in separate Kaggle sessions, runtime should be treated as descriptive rather than as a hardware-benchmark result, even though all runs report a Tesla T4.

6 Discussion

6.1 Federation is not the main source of loss in this experiment

The IID result is close to centralized training in accuracy: 77.16% versus 77.93%. The larger difference in Macro F1 (52.90% versus 56.70%) shows that the remaining cost of federation is concentrated more heavily in class-balanced behavior than in aggregate traffic-level accuracy.

This matters for IDS evaluation. A model may retain strong accuracy by performing well on common flooding classes while losing reliability on infrequent attacks. The 1.61-point Weighted F1 reduction from Centralized to FedAvg-IID is much smaller than the 3.80-point Macro F1 reduction.

6.2 Non-IID label skew is the dominant challenge

The controlled IID-to-non-IID comparison causes the largest change in every major metric. Macro F1 falls by 24.10 points and balanced accuracy by 16.33 points. The client partition statistics support the interpretation: clients are highly unequal in size, each client sees only 23.6 classes on average, and a single class occupies 59.28% of a client dataset on average.

The experiment therefore demonstrates a central FL phenomenon in a concrete IDS setting: strong local performance does not guarantee a balanced global model when client objectives are driven by very different label distributions.

6.3 Why the tested FedProx configuration may not help

FedProx is intended to stabilize optimization under heterogeneity, yet $\mu = 0.01$ produces lower Macro F1 than FedAvg here. Several experiment-specific explanations are plausible and should be tested rather than assumed:

- **Only one local epoch is used.** FedProx constrains client drift, but one local epoch already limits how far clients can move before aggregation. The benefit of a proximal term may therefore be smaller than in settings with more local computation.
- **Only one μ value is tested.** $\mu = 0.01$ may be too strong, too weak, or simply suboptimal for this model and partition.
- **The heterogeneity is severe.** Dirichlet $\alpha = 0.1$ produces both label skew and a 10.15× client-size ratio. A single global model may remain difficult to optimize with either algorithm in only three rounds.
- **The class distribution is extremely imbalanced.** The local proximal constraint does not directly address rare-class scarcity or class imbalance.

These are hypotheses motivated by the measured results. The current notebooks do not provide a μ sweep, additional local-epoch settings, or repeated seeds, so the report does not claim a causal explanation beyond the measured controlled effects.

6.4 Class imbalance is visible across every training regime

Even centralized training produces zero F1 for five classes and near-zero F1 for some reconnaissance/web labels. Because class weights are disabled and several classes have only thousands of samples in a dataset dominated by million-sample flooding classes, rare-label failure is present before federation. Non-IID partitioning then compounds that difficulty by making local class exposure even less uniform.

This suggests that any follow-up designed to improve minority-attack detection should separate two questions:

1. How much comes from *global class imbalance*?
2. How much additional loss comes from *federated client heterogeneity*?

7 Threats to Validity and Limitations

The experiment is strong as a controlled engineering comparison, but several limitations constrain research claims:

1. **Single random seed.** All runs use seed 42. There are no repeated runs, confidence intervals, or hypothesis tests, so small differences should not be interpreted as statistically stable.

2. **Short federated horizon.** Only three rounds are run. The trajectories are still improving at round 3, especially under non-IID data, so the results are not necessarily converged endpoints.

3. **One local-epoch setting.** Client drift and FedProx behavior can depend strongly on local computation. No local-epoch sensitivity study is included.

4. **One FedProx coefficient.** Only $\mu = 0.01$ is tested; there is no hyperparameter sweep.

5. **Full client participation.** All 10 clients train every round. The study does not test sampling, dropouts, stragglers, or systems heterogeneity.

6. **Single-machine simulation.** Clients are sequential virtual clients on one Tesla T4. The experiment evaluates federated optimization behavior, not distributed deployment latency, network communication, or server scalability.

7. **Centralized preprocessing statistics.** StandardScaler mean and scale are fit once on the entire training split before client partitioning. This keeps the comparison controlled but is not a strict privacy-preserving preprocessing protocol.

8. **Global test set only.** Evaluation is centralized on one shared stratified test split. Client-specific test performance and fairness across organizations are not reported.

9. **One model family.** Conclusions are specific to the 39–128–64–34 MLP and may not transfer to tree models, convolutional/recurrent models, transformers, or personalized FL methods.

10. **No class balancing.** Class weights are intentionally disabled. This isolates the current pipeline but leaves severe imbalance untreated.

11. **Communication cost not measured.** Model-update bytes, bandwidth, serialization overhead, and wall-clock network time are outside the current simulation.

8 Reproducibility and Experimental Artifacts

The supplied workflow is reproducible at the notebook level:

- the data preparation fixes the class order, train/test split, seed, and scaler;
- IID clients are deterministic and verified to cover the training set exactly once;
- non-IID clients are generated deterministically from the same training labels using $\alpha = 0.1$;
- the non-IID partition arrays are saved, and the comparison notebook verifies an exact match between FedAvg and FedProx;
- all final experiments use the same feature count, class count, architecture, batch size, learning rate, and effective data passes;
- checkpoints and CSV/JSON result artifacts are saved by each notebook.

The six analyzed notebooks are:

- `fl-dataset-preparation-markp.ipynb`
- `centralised-baseline-markp.ipynb`
- `fedavg-iid-markp.ipynb`
- `fedavg-non-iid-markp.ipynb`
- `fedprox-non-iid-markp.ipynb`
- `comparisons.ipynb`

The notebooks dynamically read the 39 feature names from the dataset header, but those names are not printed in the supplied outputs. This report therefore records the verified feature count without inventing a feature list.

9 Recommended Next Experiments

The most informative follow-up is not to add unrelated FL techniques, but to resolve the uncertainties exposed by the controlled results:

1. Extend FedAvg and FedProx to enough rounds to observe whether non-IID Macro F1 plateaus.
2. Run at least 3–5 random seeds and report mean ± standard deviation.
3. Sweep FedProx μ on a logarithmic grid, for example 0, 0.001, 0.01, 0.1, while keeping partitions fixed.
4. Repeat with 1, 2, and 5 local epochs to measure the interaction between local drift and the proximal term.
5. Add class-weighted cross-entropy or a controlled resampling strategy to distinguish class-imbalance effects from federation effects.
6. Evaluate the global model per client, not only on the centralized global test set.
7. Measure serialized update size and round communication volume before claiming systems efficiency.
8. If privacy is part of the research claim, replace globally fit preprocessing statistics with a documented local or federated preprocessing scheme.

10 Conclusion

The experiment establishes a clear controlled baseline for federated IoT intrusion detection on the full cleaned CICIoT2023 workload. The centralized MLP reaches 56.70% Macro F1. Moving to FedAvg with nearly perfectly IID clients reduces Macro F1 by 3.80 points, while preserving similar aggregate accuracy. Introducing severe Dirichlet label skew causes the dominant degradation: FedAvg Macro F1 falls by 24.10 additional points to 28.80%.

FedProx with $\mu = 0.01$ does not recover this loss under the tested three-round, one-local-epoch budget; instead Macro F1 decreases further to 23.36%. The result is valuable precisely because the comparison is controlled and negative: it shows that adding a heterogeneity-aware objective does not automatically improve a federated IDS. The next research step is therefore a sensitivity study over rounds, local epochs, μ, and random seeds, combined with explicit treatment of severe class imbalance.

A Full Class Distribution

Table 6: Class counts in the cleaned 45,018,243-row dataset.

ID	Class	Count	Share (%)
0	BACKDOOR_MALWARE	3,078	0.006837
1	BENIGN	1,051,313	2.335304
2	BROWSERHIJACKING	5,630	0.012506
3	COMMANDINJECTION	5,168	0.011480
4	DDOS-ACK_FRAGMENTATION	272,765	0.605899
5	DDOS-HTTP_FLOOD	27,597	0.061302
6	DDOS-ICMP_FLOOD	6,893,196	15.312006
7	DDOS-ICMP_FRAGMENTATION	433,115	0.962088
8	DDOS-PSHACK_FLOOD	3,920,332	8.708319
9	DDOS-RSTFINFLOOD	3,872,779	8.602688
10	DDOS-SLOWLORIS	22,399	0.049755
11	DDOS-SYNONYMOUSIP_FLOOD	3,445,626	7.653844
12	DDOS-SYN_FLOOD	3,886,050	8.632167
13	DDOS-TCP_FLOOD	4,305,990	9.564989
14	DDOS-UDP_FLOOD	5,180,969	11.508599
15	DDOS-UDP_FRAGMENTATION	274,879	0.610595
16	DICTIONARYBRUTEFORCE	12,522	0.027815
17	DNS_SPOOFING	171,463	0.380874
18	DOS-HTTP_FLOOD	68,795	0.152816
19	DOS-SYN_FLOOD	1,942,134	4.314104
20	DOS-TCP_FLOOD	2,558,194	5.682572
21	DOS-UDP_FLOOD	3,177,217	7.057621
22	MIRAI-GREETH_FLOOD	949,322	2.108750
23	MIRAI-GREIP_FLOOD	719,601	1.598465
24	MIRAI-UDPPLAIN	852,628	1.893961
25	MITM-ARPSPOOFING	294,451	0.654070
26	RECON-HOSTDISCOVERY	128,676	0.285831
27	RECON-OSSCAN	93,966	0.208729
28	RECON-PINGSWEEP	2,161	0.004800
29	RECON-PORTSCAN	78,729	0.174882
30	SQLINJECTION	5,021	0.011153
31	UPLOADING_ATTACK	1,196	0.002657
32	VULNERABILITYSCAN	357,576	0.794291
33	XSS	3,705	0.008230

B Federated Round Histories

B.1 FedAvg-IID

Round	Client loss	Client acc. (%)	Test acc. (%)	Bal. acc. (%)	Macro F1 (%)	Weighted F1 (%)	MCC
1	0.752207	71.17	74.91	45.12	45.73	72.58	0.7271
2	0.484484	75.49	76.50	50.46	50.88	74.15	0.7450
3	0.466195	76.62	77.16	52.57	52.90	74.99	0.7521

B.2 FedAvg-Non-IID

Round	Client loss	Client acc. (%)	Test acc. (%)	Bal. acc. (%)	Macro F1 (%)	Weighted F1 (%)	MCC
1	0.279074	94.73	61.63	28.82	21.59	51.89	0.6007
2	0.133591	95.79	62.82	32.18	24.25	54.85	0.6104
3	0.119808	95.99	65.74	36.24	28.80	57.91	0.6336

B.3 FedProx-Non-IID

Round	CE loss	Objective	Prox. penalty	Client acc. (%)	Test acc. (%)	Bal. acc. (%)	Macro F1 (%)	Weighted F1 (%)	MC
1	0.297907	0.381191	0.083284	94.53	62.48	28.88	21.26	53.95	0.60
2	0.159411	0.190106	0.030695	95.75	62.93	30.51	22.09	54.78	0.61
3	0.128508	0.143329	0.014821	95.89	64.91	31.28	23.36	56.95	0.62

C Full Per-Class Precision, Recall, and F1

C.1 Centralized

Table 7: Per-class metrics: Centralized.

ID	Class	Support	Precision	Recall	F1
0	BACKDOOR_MALWARE	616	0.000000	0.000000	0.000000
1	BENIGN	210,263	0.723957	0.911183	0.806851
2	BROWSERHIJACKING	1,126	0.888889	0.056838	0.106845
3	COMMANDINJECTION	1,034	0.825503	0.118956	0.207946
4	DDOS-ACK_FRAGMENTATION	54,553	0.997566	0.984089	0.990781
5	DDOS-HTTP_FLOOD	5,519	0.872977	0.645044	0.741899
6	DDOS-ICMP_FLOOD	1,378,639	0.998591	0.999189	0.998890
7	DDOS-ICMP_FRAGMENTATION	86,623	0.996798	0.981010	0.988841
8	DDOS-PSHACK_FLOOD	784,067	0.999731	0.998369	0.999049
9	DDOS-RSTFINFLOOD	774,556	0.999503	0.999129	0.999316
10	DDOS-SLOWLORIS	4,480	0.735326	0.746652	0.740946
11	DDOS-SYNONYMOUSIP_FLOOD	689,125	0.468289	0.630932	0.537578
12	DDOS-SYN_FLOOD	777,210	0.486887	0.480803	0.483826
13	DDOS-TCP_FLOOD	861,198	0.682487	0.876494	0.767419
14	DDOS-UDP_FLOOD	1,036,194	0.726176	0.943846	0.820826
15	DDOS-UDP_FRAGMENTATION	54,976	0.989014	0.985776	0.987392
16	DICTIONARYBRUTEFORCE	2,504	0.987981	0.164137	0.281507

ID	Class	Support	Precision	Recall	F1
17	DNS_SPOOFING	34,293	0.729847	0.400519	0.517209
18	DOS-HTTP_FLOOD	13,759	0.834648	0.922669	0.876454
19	DOS-SYN_FLOOD	388,427	0.605342	0.247055	0.350900
20	DOS-TCP_FLOOD	511,639	0.602589	0.313385	0.412331
21	DOS-UDP_FLOOD	635,443	0.816364	0.421286	0.555767
22	MIRAI-GREETH_FLOOD	189,864	0.993298	0.991389	0.992342
23	MIRAI-GREIP_FLOOD	143,920	0.988250	0.991718	0.989981
24	MIRAI-UDPPLAIN	170,526	0.995292	0.996681	0.995986
25	MITM-ARPSPOOFING	58,890	0.700511	0.672474	0.686206
26	RECON-HOSTDISCOVERY	25,735	0.619171	0.595376	0.607040
27	RECON-OSSCAN	18,793	0.395927	0.201724	0.267273
28	RECON-PINGSWEEP	432	0.000000	0.000000	0.000000
29	RECON-PORTSCAN	15,746	0.203980	0.002604	0.005142
30	SQLINJECTION	1,004	0.000000	0.000000	0.000000
31	UPLOADING_ATTACK	239	0.000000	0.000000	0.000000
32	VULNERABILITYSCAN	71,515	0.575109	0.547494	0.560962
33	XSS	741	0.000000	0.000000	0.000000

C.2 FedAvg-IID

Table 8: Per-class metrics: FedAvg-IID.

ID	Class	Support	Precision	Recall	F1
0	BACKDOOR_MALWARE	616	0.000000	0.000000	0.000000
1	BENIGN	210,263	0.677936	0.898751	0.772881
2	BROWSERHIJACKING	1,126	0.000000	0.000000	0.000000
3	COMMANDINJECTION	1,034	0.000000	0.000000	0.000000
4	DDOS-ACK_FRAGMENTATION	54,553	0.988718	0.983117	0.985910
5	DDOS-HTTP_FLOOD	5,519	0.709265	0.563145	0.627815
6	DDOS-ICMP_FLOOD	1,378,639	0.999472	0.998911	0.999191
7	DDOS-ICMP_FRAGMENTATION	86,623	0.976499	0.983826	0.980149
8	DDOS-PSHACK_FLOOD	784,067	0.999302	0.998305	0.998803
9	DDOS-RSTFINFLOOD	774,556	0.999862	0.998989	0.999425
10	DDOS-SLOWLORIS	4,480	0.479526	0.677009	0.561407
11	DDOS-SYNONYMOUSIP_FLOOD	689,125	0.468380	0.601132	0.526517
12	DDOS-SYN_FLOOD	777,210	0.472681	0.523050	0.496592
13	DDOS-TCP_FLOOD	861,198	0.653956	0.948871	0.774282
14	DDOS-UDP_FLOOD	1,036,194	0.722883	0.946015	0.819533
15	DDOS-UDP_FRAGMENTATION	54,976	0.986226	0.982010	0.984114
16	DICTIONARYBRUTEFORCE	2,504	0.831386	0.198882	0.320980
17	DNS_SPOOFING	34,293	0.577643	0.281486	0.378519
18	DOS-HTTP_FLOOD	13,759	0.784024	0.895995	0.836279
19	DOS-SYN_FLOOD	388,427	0.644840	0.180312	0.281820

ID	Class	Support	Precision	Recall	F1
20	DOS-TCP_FLOOD	511,639	0.643400	0.153626	0.248029
21	DOS-UDP_FLOOD	635,443	0.820235	0.408997	0.545826
22	MIRAI-GREETH_FLOOD	189,864	0.971330	0.971790	0.971560
23	MIRAI-GREIP_FLOOD	143,920	0.962938	0.957358	0.960140
24	MIRAI-UDPPLAIN	170,526	0.990883	0.994886	0.992881
25	MITM-ARPSPOOFING	58,890	0.674514	0.519154	0.586724
26	RECON-HOSTDISCOVERY	25,735	0.597405	0.522401	0.557391
27	RECON-OSSCAN	18,793	0.374212	0.028415	0.052819
28	RECON-PINGSWEEP	432	0.000000	0.000000	0.000000
29	RECON-PORTSCAN	15,746	0.492778	0.132161	0.208423
30	SQLINJECTION	1,004	0.000000	0.000000	0.000000
31	UPLOADING_ATTACK	239	0.000000	0.000000	0.000000
32	VULNERABILITYSCAN	71,515	0.506528	0.526743	0.516438
33	XSS	741	0.000000	0.000000	0.000000

C.3 FedAvg-Non-IID

Table 9: Per-class metrics: FedAvg-Non-IID.

ID	Class	Support	Precision	Recall	F1
0	BACKDOOR_MALWARE	616	0.000000	0.000000	0.000000
1	BENIGN	210,263	0.000000	0.000000	0.000000
2	BROWSERHIJACKING	1,126	0.000000	0.000000	0.000000
3	COMMANDINJECTION	1,034	0.000000	0.000000	0.000000
4	DDOS-ACK_FRAGMENTATION	54,553	0.829698	0.988085	0.901991
5	DDOS-HTTP_FLOOD	5,519	0.524751	0.562783	0.543102
6	DDOS-ICMP_FLOOD	1,378,639	0.999549	0.998757	0.999153
7	DDOS-ICMP_FRAGMENTATION	86,623	1.000000	0.278044	0.435109
8	DDOS-PSHACK_FLOOD	784,067	0.993073	0.997485	0.995274
9	DDOS-RSTFINFLOOD	774,556	0.999835	0.999024	0.999429
10	DDOS-SLOWLORIS	4,480	0.000000	0.000000	0.000000
11	DDOS-SYNONYMOUSIP_FLOOD	689,125	0.454545	0.000109	0.000218
12	DDOS-SYN_FLOOD	777,210	0.429146	0.606122	0.502507
13	DDOS-TCP_FLOOD	861,198	0.628563	0.998855	0.771582
14	DDOS-UDP_FLOOD	1,036,194	0.613936	0.998502	0.760360
15	DDOS-UDP_FRAGMENTATION	54,976	0.202615	0.985485	0.336123
16	DICTIONARYBRUTEFORCE	2,504	0.120303	0.310703	0.173448
17	DNS_SPOOFING	34,293	0.104719	0.951769	0.188678
18	DOS-HTTP_FLOOD	13,759	0.000000	0.000000	0.000000
19	DOS-SYN_FLOOD	388,427	0.291901	0.581270	0.388637
20	DOS-TCP_FLOOD	511,639	0.000000	0.000000	0.000000
21	DOS-UDP_FLOOD	635,443	0.000000	0.000000	0.000000
22	MIRAI-GREETH_FLOOD	189,864	0.567842	0.995560	0.723193

ID	Class	Support	Precision	Recall	F1
23	MIRAI-GREIP_FLOOD	143,920	0.048387	0.000104	0.000208
24	MIRAI-UDPPLAIN	170,526	0.000000	0.000000	0.000000
25	MITM-ARPSPOOFING	58,890	0.229181	0.334284	0.271930
26	RECON-HOSTDISCOVERY	25,735	0.611157	0.456344	0.522525
27	RECON-OSSCAN	18,793	0.000000	0.000000	0.000000
28	RECON-PINGSWEEP	432	0.000000	0.000000	0.000000
29	RECON-PORTSCAN	15,746	0.230343	0.261590	0.244974
30	SQLINJECTION	1,004	0.000000	0.000000	0.000000
31	UPLOADING_ATTACK	239	0.000000	0.000000	0.000000
32	VULNERABILITYSCAN	71,515	0.436548	0.016836	0.032421
33	XSS	741	0.000000	0.000000	0.000000

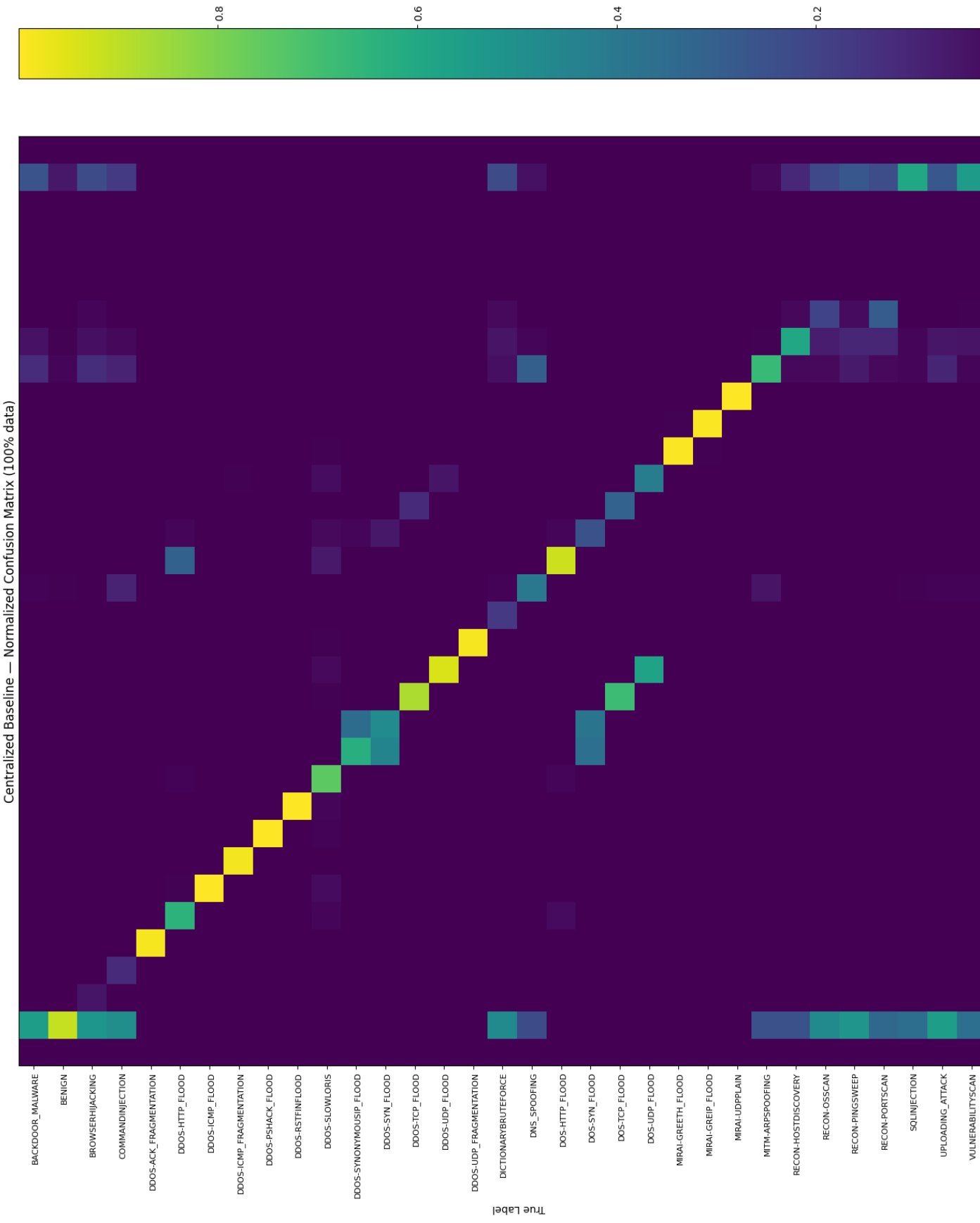
C.4 FedProx-Non-IID

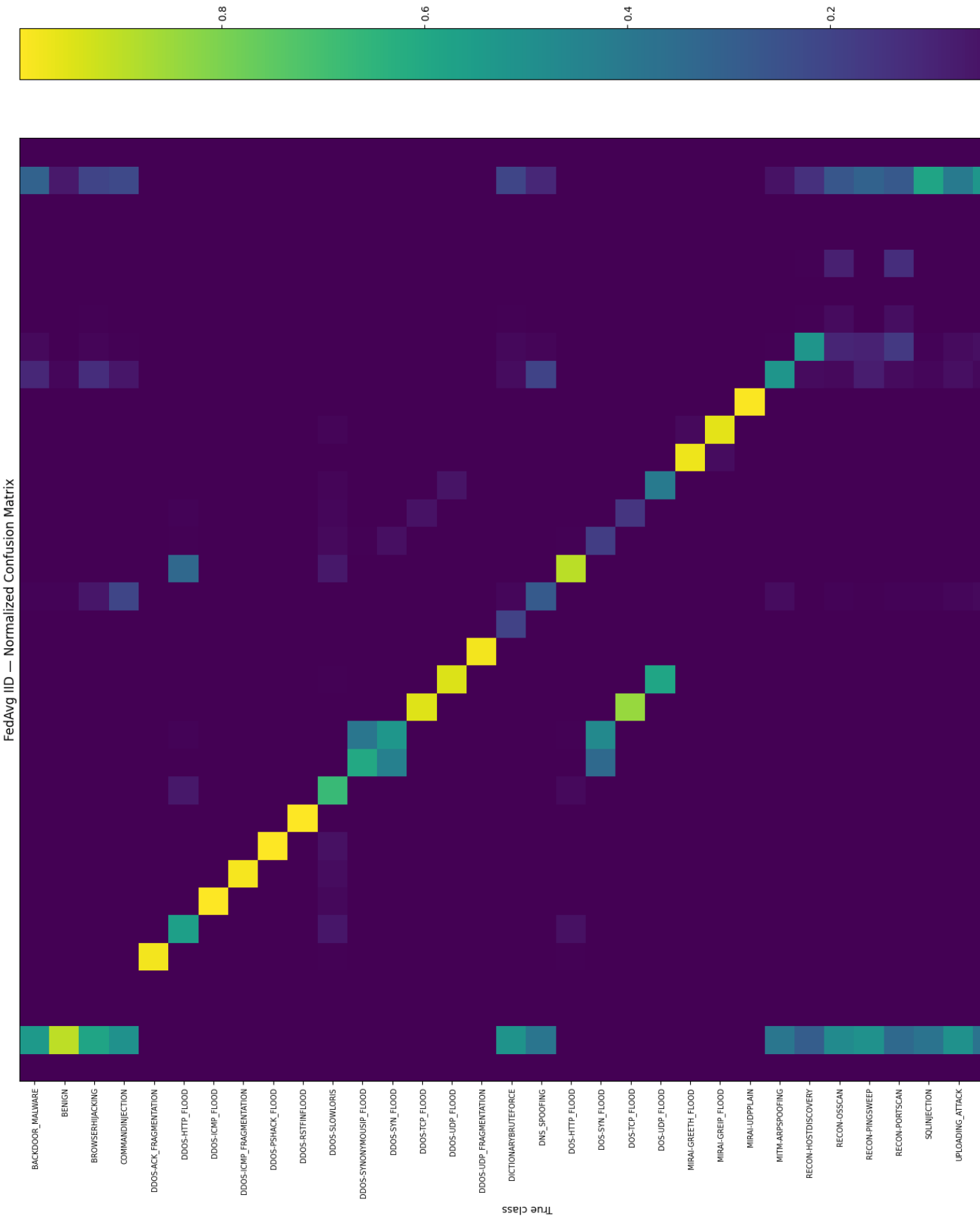
Table 10: Per-class metrics: FedProx-Non-IID.

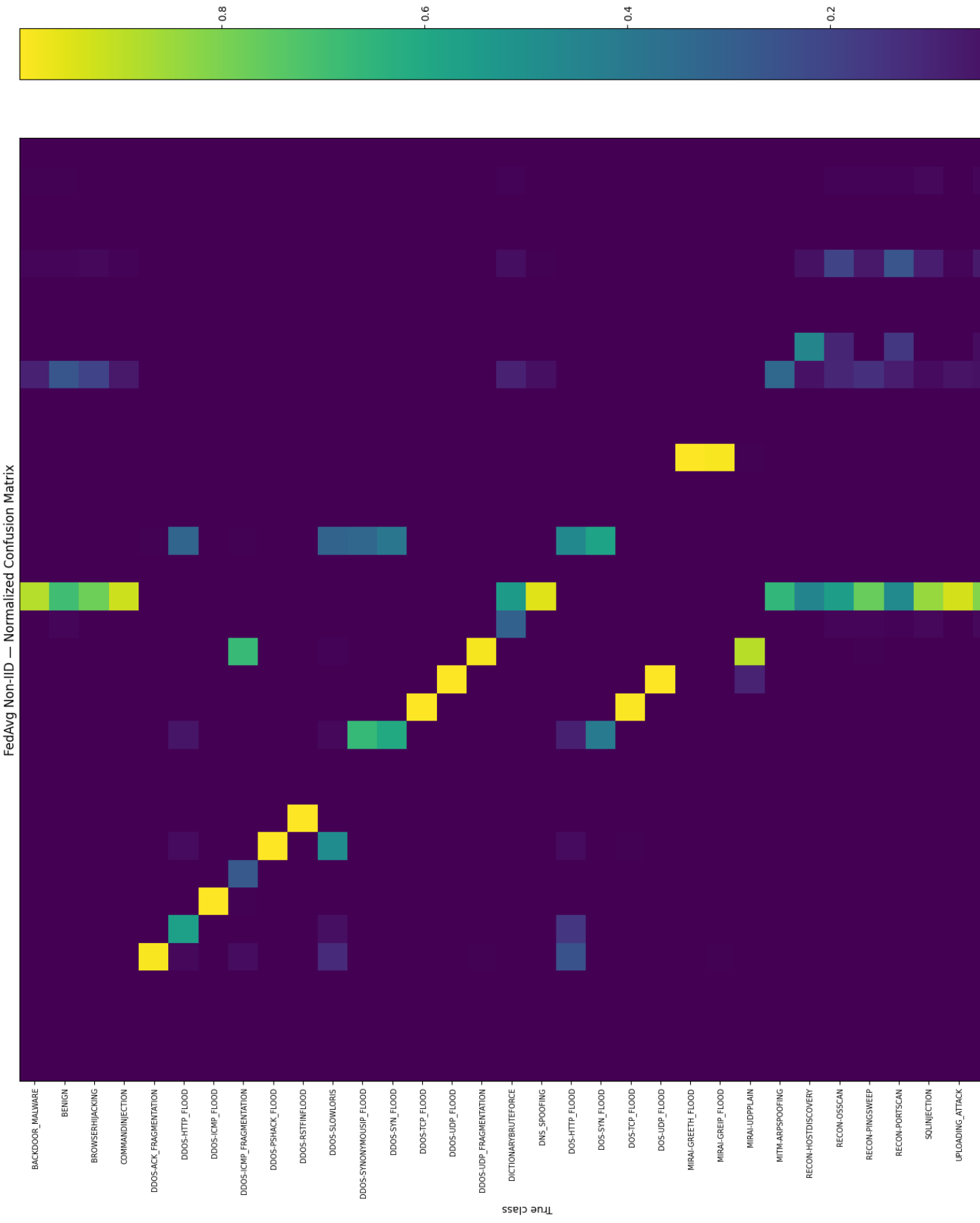
ID	Class	Support	Precision	Recall	F1
0	BACKDOOR_MALWARE	616	0.000000	0.000000	0.000000
1	BENIGN	210,263	0.000000	0.000000	0.000000
2	BROWSERHIJACKING	1,126	0.000000	0.000000	0.000000
3	COMMANDINJECTION	1,034	0.000000	0.000000	0.000000
4	DDOS-ACK_FRAGMENTATION	54,553	0.759707	0.984877	0.857761
5	DDOS-HTTP_FLOOD	5,519	0.314815	0.003080	0.006101
6	DDOS-ICMP_FLOOD	1,378,639	0.997724	0.998895	0.998309
7	DDOS-ICMP_FRAGMENTATION	86,623	0.000000	0.000000	0.000000
8	DDOS-PSHACK_FLOOD	784,067	0.981245	0.994220	0.987690
9	DDOS-RSTFINFLOOD	774,556	0.990734	0.998754	0.994728
10	DDOS-SLOWLORIS	4,480	0.000000	0.000000	0.000000
11	DDOS-SYNONYMOUSIP_FLOOD	689,125	0.395277	0.001117	0.002228
12	DDOS-SYN_FLOOD	777,210	0.433920	0.486817	0.458849
13	DDOS-TCP_FLOOD	861,198	0.628704	0.998811	0.771675
14	DDOS-UDP_FLOOD	1,036,194	0.618219	0.998907	0.763754
15	DDOS-UDP_FRAGMENTATION	54,976	0.171394	0.982920	0.291891
16	DICTIONARYBRUTEFORCE	2,504	0.500000	0.000799	0.001595
17	DNS_SPOOFING	34,293	0.113348	0.790599	0.198270
18	DOS-HTTP_FLOOD	13,759	0.000000	0.000000	0.000000
19	DOS-SYN_FLOOD	388,427	0.279774	0.705226	0.400617
20	DOS-TCP_FLOOD	511,639	0.000000	0.000000	0.000000
21	DOS-UDP_FLOOD	635,443	0.050000	0.000003	0.000006
22	MIRAI-GREETH_FLOOD	189,864	0.569078	0.995565	0.724196
23	MIRAI-GREIP_FLOOD	143,920	0.071429	0.000292	0.000581
24	MIRAI-UDPPLAIN	170,526	0.000000	0.000000	0.000000
25	MITM-ARPSPOOFING	58,890	0.200511	0.574427	0.297260

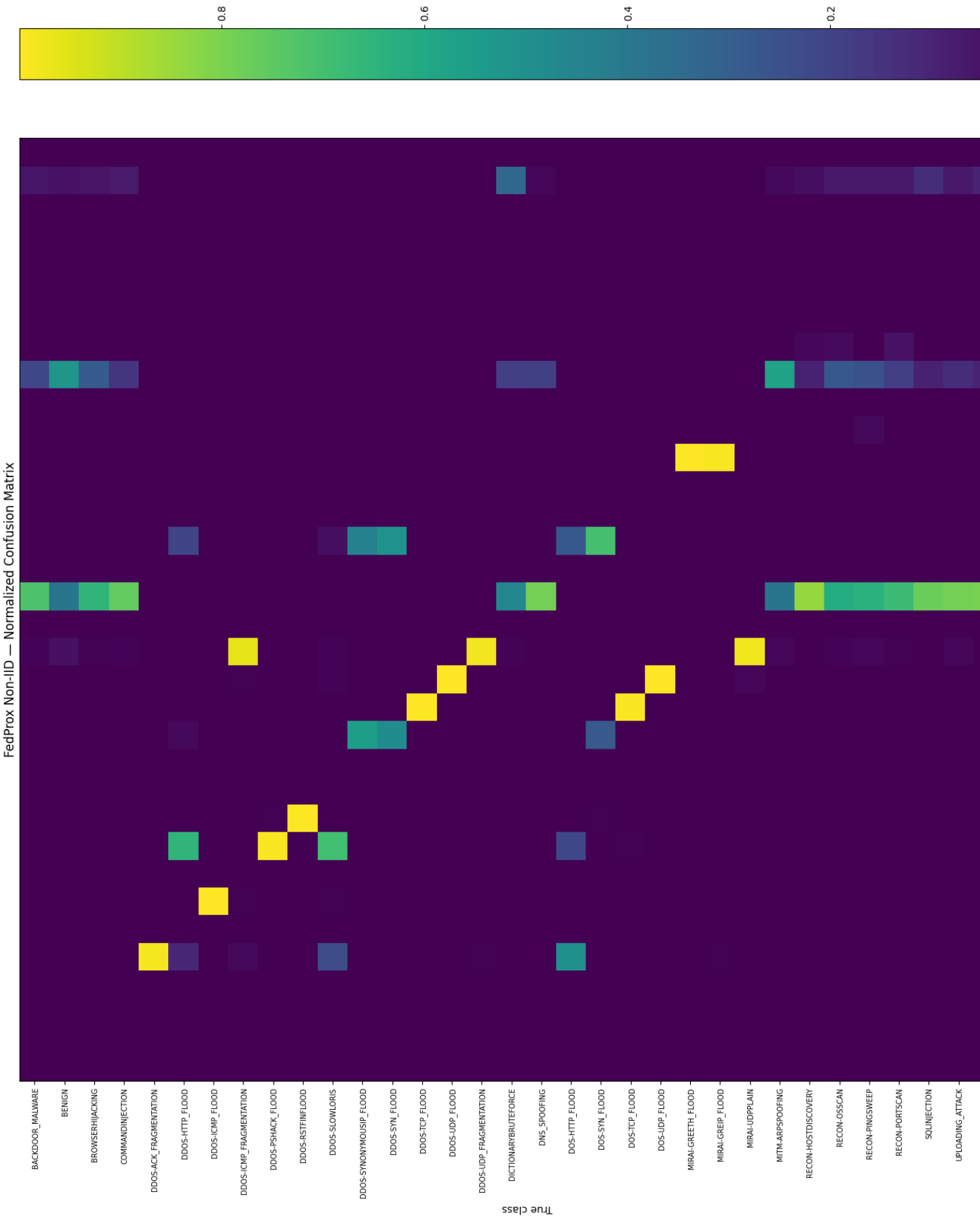
ID	Class	Support	Precision	Recall	F1
26	RECON-HOSTDISCOVERY	25,735	0.263246	0.018341	0.034292
27	RECON-OSSCAN	18,793	0.000000	0.000000	0.000000
28	RECON-PINGSWEEP	432	0.000000	0.000000	0.000000
29	RECON-PORTSCAN	15,746	0.251852	0.002159	0.004282
30	SQLINJECTION	1,004	0.000000	0.000000	0.000000
31	UPLOADING_ATTACK	239	0.000000	0.000000	0.000000
32	VULNERABILITYSCAN	71,515	0.295578	0.099252	0.148604
33	XSS	741	0.000000	0.000000	0.000000

D Normalized Confusion Matrices


Centralized Baseline — Normalized Confusion Matrix (100% data)
True Label
0.8
0.6
0.4
0.2
BACKDOOR_MALWARE
BENIGN
BROWSERHIJACKING
COMMANDINJECTION
DDOS-ACK_FRAGMENTATION
DDOS-HTTP_FLOOD
DDOS-ICMP_FLOOD
DDOS-ICMP_FRAGMENTATION
DDOS-PSHACK_FLOOD
DDOS-RSTFINFLOOD
DDOS-SLOWLORIS
DDOS-SYNONYMOUSIP_FLOOD
DDOS-SYN_FLOOD
DDOS-TCP_FLOOD
DDOS-UDP_FLOOD
DDOS-UDP_FRAGMENTATION
DICTIONARYBRUTEFORCE
DNS_SPOOFING
DOS-HTTP_FLOOD
DOS-SYN_FLOOD
DOS-TCP_FLOOD
DOS-UDP_FLOOD
MIRAI-GREETH_FLOOD
MIRAI-GREIP_FLOOD
MIRAI-UDPPLAIN
MITM-ARPSPOOFING
RECON-HOSTDISCOVERY
RECON-OSSCAN
RECON-PINGSWEEP
RECON-PORTSCAN
SQLINJECTION
UPLOADING_ATTACK
VULNERABILITYSCAN


FedAvg IID — Normalized Confusion Matrix
True class
0.8
0.6
0.4
0.2
BACKDOOR_MALWARE
BENIGN
BROWSERHIJACKING
COMMANDINJECTION
DDOS-ACK_FRAGMENTATION
DDOS-HTTP_FLOOD
DDOS-ICMP_FLOOD
DDOS-ICMP_FRAGMENTATION
DDOS-PSHACK_FLOOD
DDOS-RSTFINFLOOD
DDOS-SLOWLORIS
DDOS-SYNONYMOUSIP_FLOOD
DDOS-SYN_FLOOD
DDOS-TCP_FLOOD
DDOS-UDP_FLOOD
DDOS-UDP_FRAGMENTATION
DICTIONARYBRUTEFORCE
DNS_SPOOFING
DOS-HTTP_FLOOD
DOS-SYN_FLOOD
DOS-TCP_FLOOD
DOS-UDP_FLOOD
MIRAI-GREETH_FLOOD
MIRAI-GREIP_FLOOD
MIRAI-UDPPLAIN
MITM-ARPSPOOFING
RECON-HOSTDISCOVERY
RECON-OSSCAN
RECON-PINGSWEEP
RECON-PORTSCAN
SQLINJECTION
UPLOADING_ATTACK


FedAvg Non-IID — Normalized Confusion Matrix
True class
0.8
0.6
0.4
0.2
BACKDOOR_MALWARE
BENIGN
BROWSERHIJACKING
COMMANDINJECTION
DDOS-ACK_FRAGMENTATION
DDOS-HTTP_FLOOD
DDOS-ICMP_FLOOD
DDOS-ICMP_FRAGMENTATION
DDOS-PSHACK_FLOOD
DDOS-RSTFINFLOOD
DDOS-SLOWLORIS
DDOS-SYNONYMOUSIP_FLOOD
DDOS-SYN_FLOOD
DDOS-TCP_FLOOD
DDOS-UDP_FLOOD
DDOS-UDP_FRAGMENTATION
DICTIONARYBRUTEFORCE
DNS_SPOOFING
DOS-HTTP_FLOOD
DOS-SYN_FLOOD
DOS-TCP_FLOOD
DOS-UDP_FLOOD
MIRAI-GREETH_FLOOD
MIRAI-GREIP_FLOOD
MIRAI-UDPPLAIN
MITM-ARPSPOOFING
RECON-HOSTDISCOVERY
RECON-OSSCAN
RECON-PINGSWEEP
RECON-PORTSCAN
SQLINJECTION
UPLOADING_ATTACK


FedProx Non-IID — Normalized Confusion Matrix
True class
0.8
0.6
0.4
0.2
BACKDOOR_MALWARE
BENIGN
BROWSERHIJACKING
COMMANDINJECTION
DDOS-ACK_FRAGMENTATION
DDOS-HTTP_FLOOD
DDOS-ICMP_FLOOD
DDOS-ICMP_FRAGMENTATION
DDOS-PSHACK_FLOOD
DDOS-RSTFINFLOOD
DDOS-SLOWLORIS
DDOS-SYNONYMOUSIP_FLOOD
DDOS-SYN_FLOOD
DDOS-TCP_FLOOD
DDOS-UDP_FLOOD
DDOS-UDP_FRAGMENTATION
DICTIONARYBRUTEFORCE
DNS_SPOOFING
DOS-HTTP_FLOOD
DOS-SYN_FLOOD
DOS-TCP_FLOOD
DOS-UDP_FLOOD
MIRAI-GREETH_FLOOD
MIRAI-GREIP_FLOOD
MIRAI-UDPPLAIN
MITM-ARPSPOOFING
RECON-HOSTDISCOVERY
RECON-OSSCAN
RECON-PINGSWEEP
RECON-PORTSCAN
SQLINJECTION
UPLOADING_ATTACK

References

[1] E. C. P. Neto, S. Dadkhah, R. Ferreira, A. Zohourian, R. Lu, and A. A. Ghorbani, "CICIoT2023: A Real-Time Dataset and Benchmark for Large-Scale Attacks in IoT Environment," *Sensors*, vol. 23, no. 13, 5941, 2023. doi: 10.3390/s23135941.

[2] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. Agüera y Arcas, "Communication-Efficient Learning of Deep Networks from Decentralized Data," in *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*, PMLR 54, pp. 1273–1282, 2017. https://proceedings.mlr.press/v54/mcmahan17a.html.

[3] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated Optimization in Heterogeneous Networks," *Proceedings of Machine Learning and Systems*, vol. 2, 2020. https://proceedings.mlsys.org/paper/2020/hash/1f5fe83998a09396ebe6477d9475ba0c-Abstract.html.